



Product Training

VSLAM Acquisition

Voyis VSLAM powered by EIVA NaviSuite V1.3

Version: F

Proprietary & Confidential



WELCOME

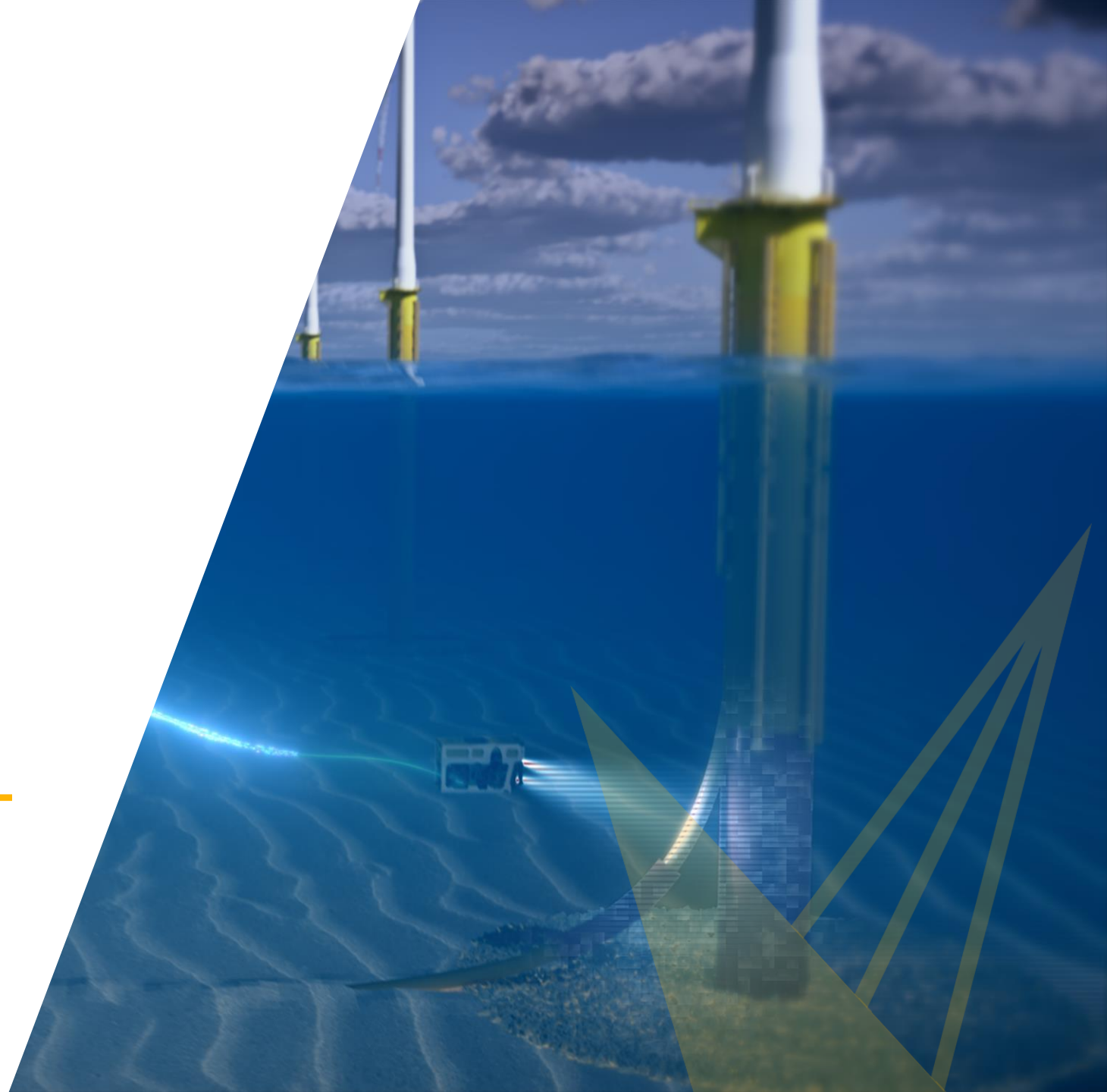
Content

1. What is VOYIS VSLAM?
2. System Requirements
3. Principles of Operation
4. System Control
5. Maintenance & Care
6. Best Practices



Tim Scott

Voyis Technical Support



What is VSLAM software?





OVERVIEW

What is VSLAM?

Voyis VSLAM powered by EIVA NaviSuite :

- Built on the proven EIVA NaviSuite platform
- Tight integration with the Voyis Discovery Stereo

It provides an **acquisition interface** for the Discovery Stereo vision system.
Which includes:

1. Camera Controls & Status Information
2. Low-Latency Video Viewer
3. Real-time 3D point cloud generation





Purpose

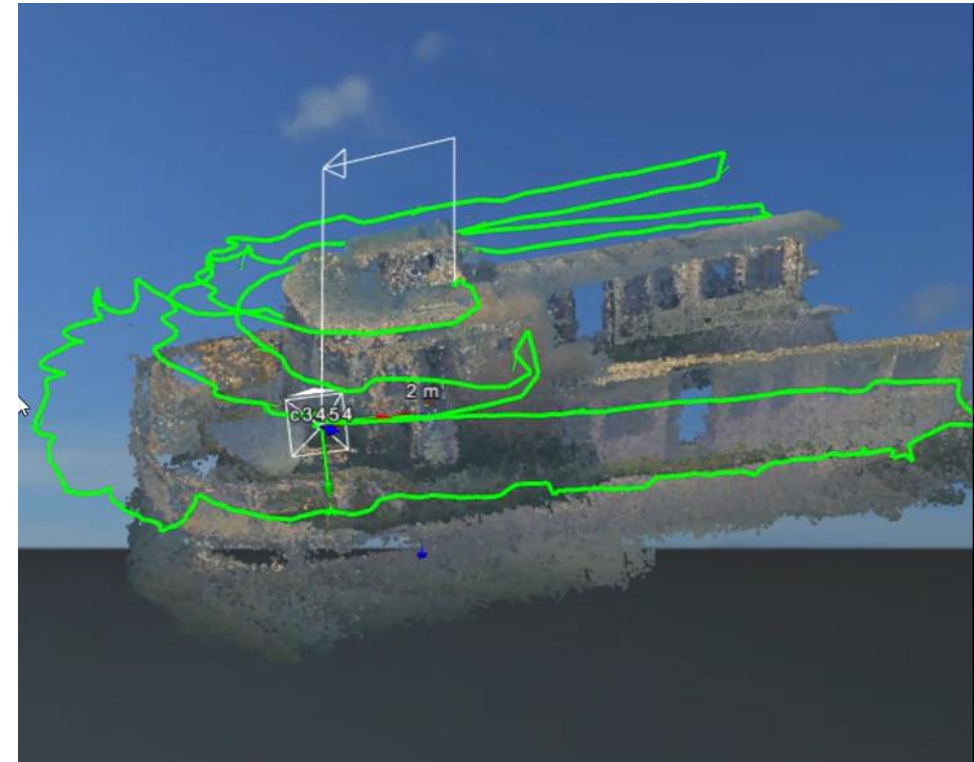
The purpose of VSLAM is to provide **Survey Feedback** to the operator to ensure effective data is being collected for photogrammetry

1. Target Coverage Feedback

- Real-time point cloud displays 3D data on areas that have been surveyed

2. Data Quality Validation

- A high-quality real-time 3D model validates that the images captured have sufficient features for tracking & modelling

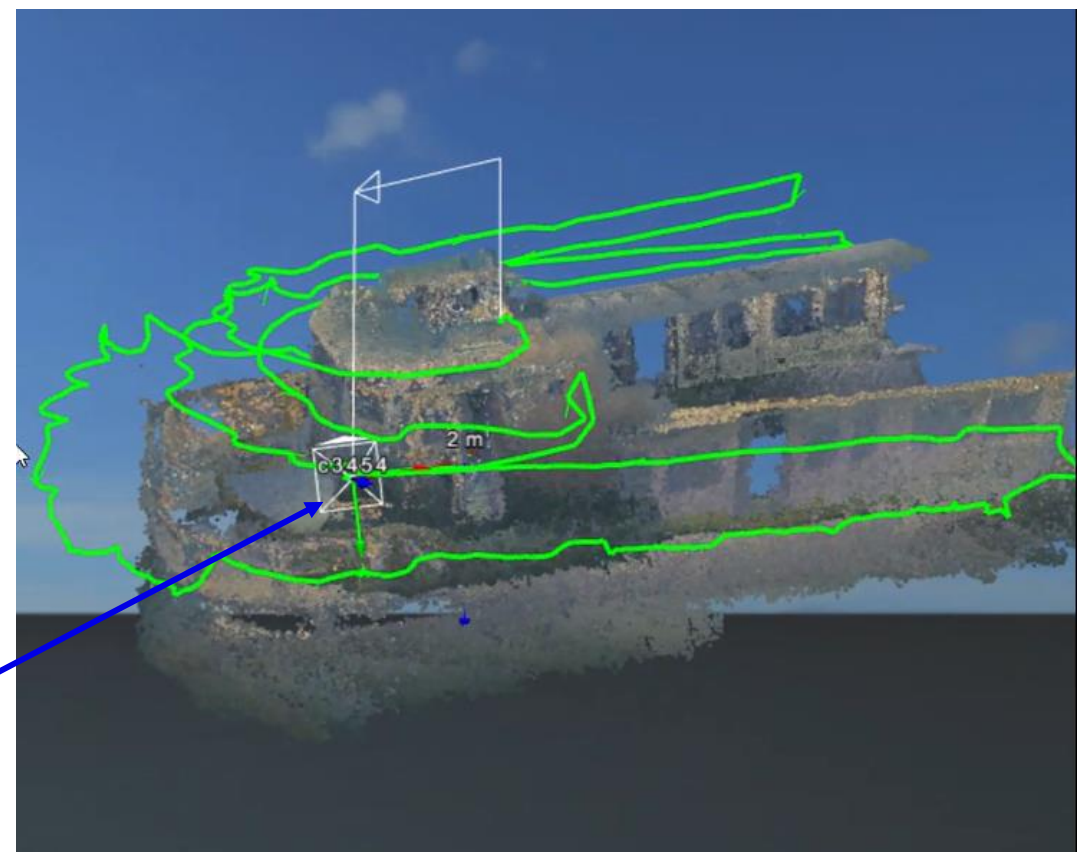




VSLAM Definitions

Visual Simultaneous **Localization** and **Mapping** (VSLAM)

- **Visual:** Utilizes stereo camera images to model and track features in the environment.
- **Localization:** Accurately tracks the **position** and orientation of the camera or vehicle within the environment.
- **Mapping:** Creates a real-time 3D point cloud of the surroundings as the camera moves through space.



System Requirements





SYSTEM REQUIREMENTS

Discovery Stereo



Discovery Stereo & Nova
Mini – 1000m & 4000m



Discovery Stereo & Nova Mini – 300m

VSLAM only works with the Voyis Discovery Stereo products.

Supports the inspection of vertical targets and enhances general visual inspections.

Simultaneously delivers video, stills images, and 3D Data



Hardware Compatibility

To successfully run the software, your computer must have a minimum of:

- **CPU:** Intel® Core™ i7-10875H, CPU @ 2.30 GHz or similar
- **RAM:** 32 GB; 64GB is preferred
- **GPU:** NVIDIA GPU with a minimum CUDA® Compute Capability of **7.5** ([link](#))
 - GPU RAM: 4GB minimum
 - Ex. Nvidia RTX 4000 Ada is an acceptable Laptop GPU
- **Disk Space:** 1 TB (depending on the survey size).
- **Network:** Ethernet connection with 500 Mbps bandwidth.



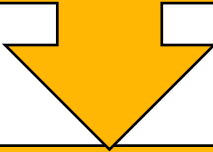
SYSTEM REQUIREMENTS

Software Compatibility

Software Versions

VSLAM Software → Newest Version (1.3)

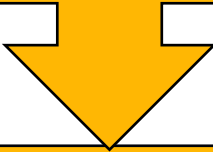
Voyis Discovery Stereo → Firmware version above (2.0.1)



Licensing

Install the software

Activate License: Help → License Activator.



Update Computer's Nvidia GPU

Ensure that NVIDIA CUDA version 12.4 is installed



Getting Started





GETTING STARTED

Software Installation

- Once you have confirmed your computer can run Voyis VSLAM, you are now ready to install the latest version of the software.
- Visit EIVA's website to download the software. The website can be found below:

<https://download.eiva.com/>



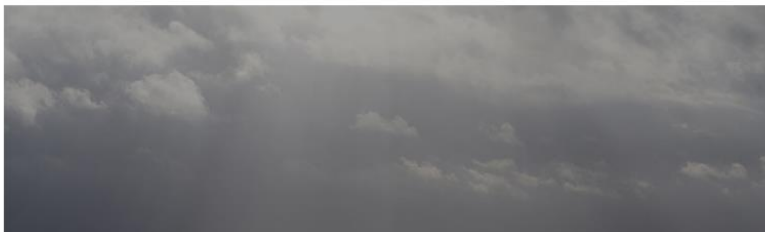
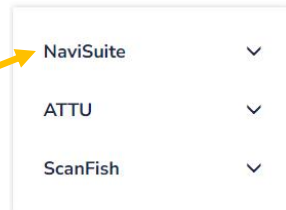
Download Site

Welcome to the EIVA Download Site, which allows you to easily download the newest version of your EIVA software solutions as well as access firmware for your EIVA hardware solutions.

Download the [NaviSuite Licence Activator](#) to activate new subscription licences on this computer or to create licence files for activation on other computers.

If you have any questions or comments to the content, please do not hesitate to contact us at support@eiva.com or +45 2048 7737.

Click on NaviSuite and
scroll to the bottom to find
the VOYIS VSLAM Software





GETTING STARTED

Software Installation

- Visit EIVA's website to download the software. The website can be found below:

<https://download.eiva.com/>

Download latest release

Name/Version		Date			
	Voyis VSLAM Powered by EIVA NaviSuite – 1.3	28 July 2025	Release notes	871 MB	Download

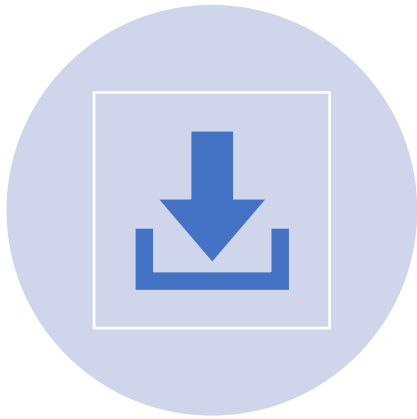
Click on download. Once your files have been downloaded you will need to extract them to your computer.



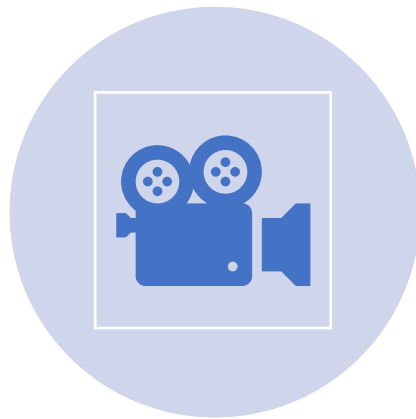
GETTING STARTED

Software Installation

- For a detailed Installation Guide – Visit EIVA’s website: [Link](#)

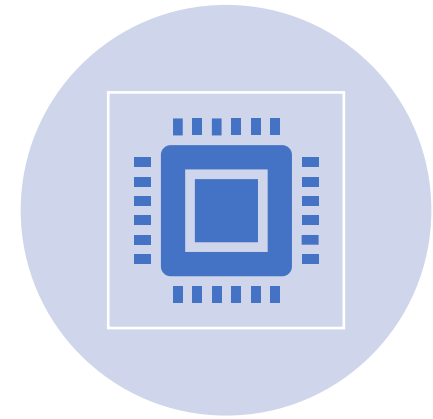


Once downloaded, you will find two .Exe files you will need to install to run VSLAM.



Discovery Powered by NaviSuite: this is the graphical user interface (GUI) program that you will interact with.

 `DiscoveryPoweredByNaviSuite_1.0.0.exe`



NaviSuite VSLAM: This background service processes the images to generate 3D Point Clouds

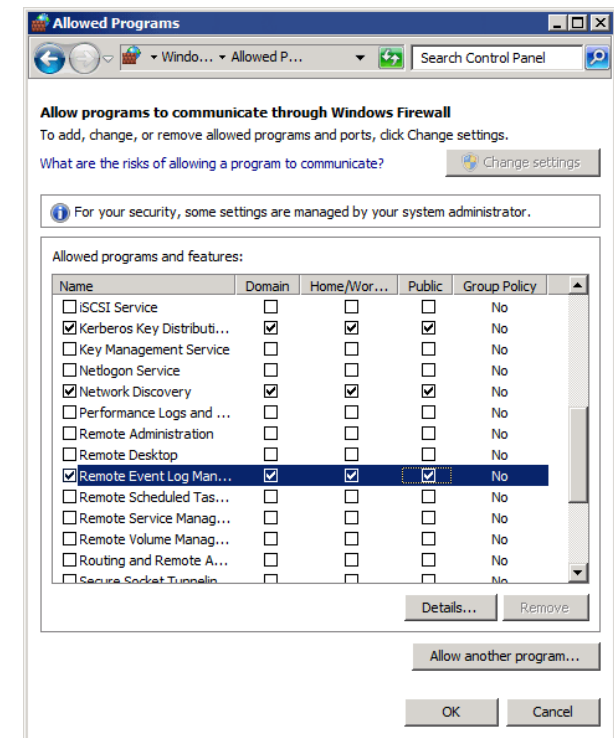
 `NaviSuiteVSLAM_1.0.0.exe`



Software Installation – Firewalls

Before running the program, it is recommended to enable all Windows firewall permissions for both Discovery Powered by NaviSuite and NaviSuite VSLAM.

1. Click **Start > Control Panel > System and Security** and click **Allow a program through Windows Firewall**, under Windows Firewall category.
2. From **Allowed programs and features** list, enable the following rules:
 - Remote Event Log Management
 - File and Printer Sharing
 - Network Discovery
3. Select **Domain**, **Private** and **Public** for each rule mentioned above.
4. Click **OK**.



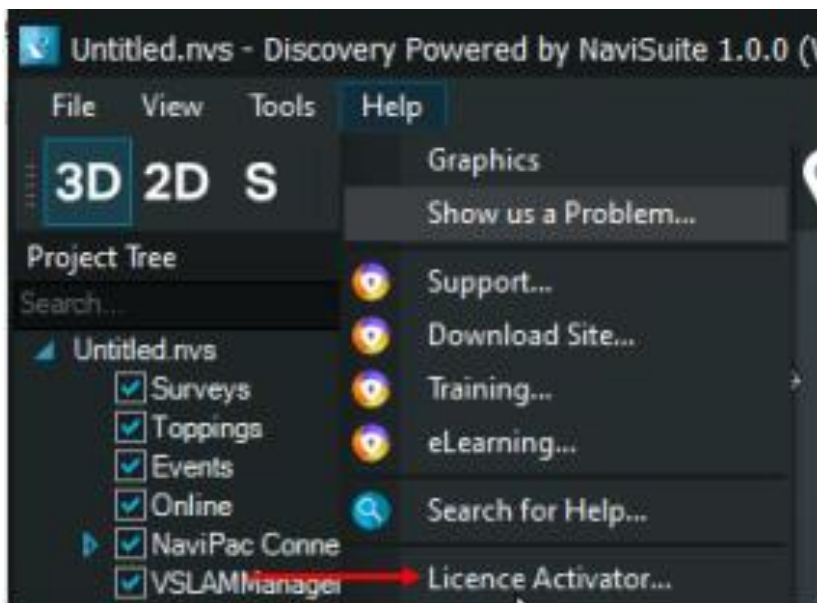
***It is recommended to restart your computer after you have enabled the permissions.
After restart, check that the firewall permissions are still enabled***



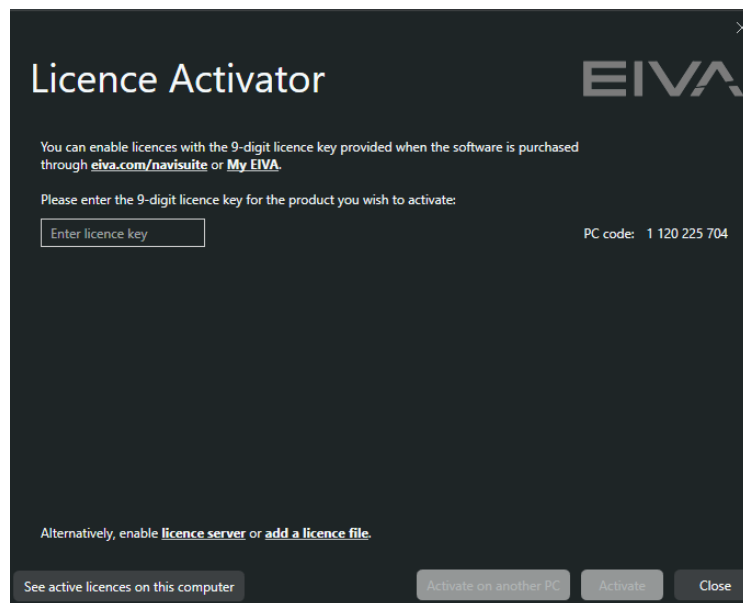
GETTING STARTED

License Activation

- The license needs to be activated in the VSLAM software.



Select **Help>License Activator**



Add your 9-digit key



Check the license is activated **Help > About**



GETTING STARTED

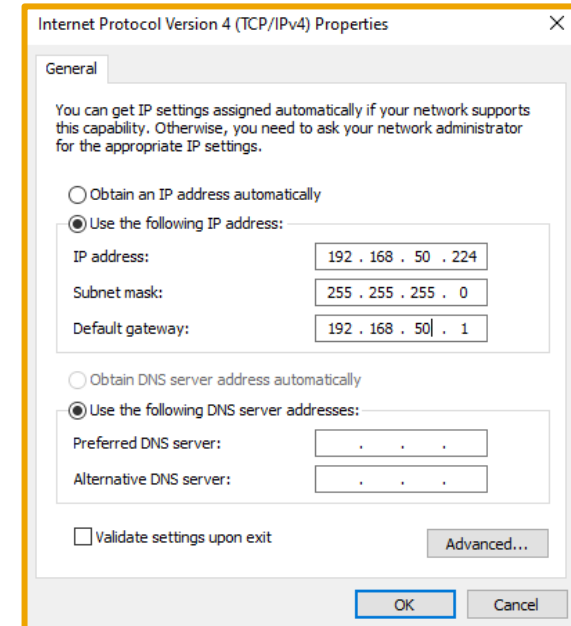
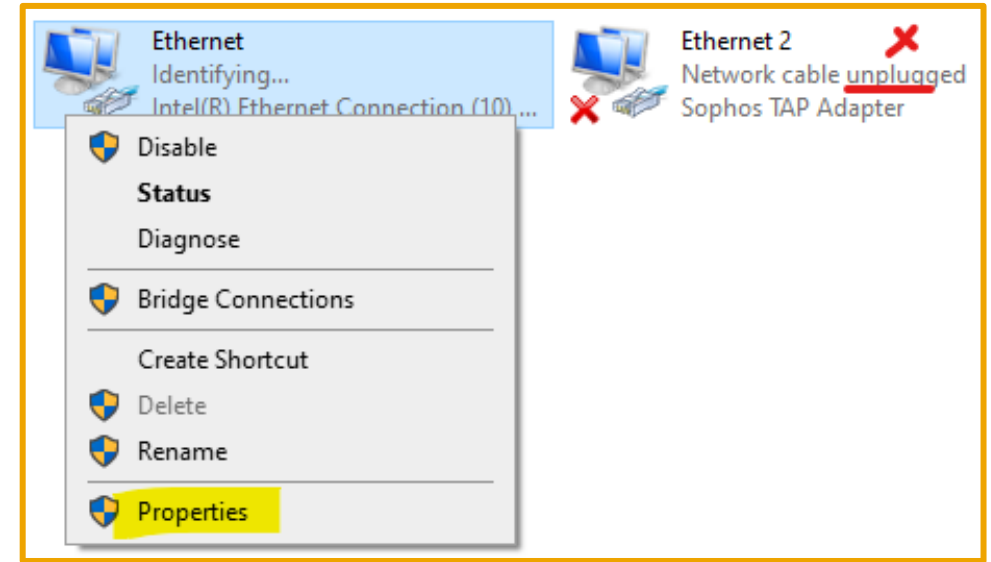
Camera Connection

(1) Setup

- Ensure Discovery has 24 VDC power
- Ensure control computer has Ethernet connection to Discovery
- Discovery GUI installed on control computer

(2) IP Settings

- Adjust your computer's subnet to match the Discovery Camera subnet found on the product serial label on camera housing.
 - From the start menu, navigate to Ethernet Settings and choose **Change Adapter Options**.
 - Select **“Properties”** on the Ethernet Connection connected to the Discovery Product.
 - “Use the following IP Address”
 - Product IP Address: 192.168.50.xxx (xxx = any number from 2 to 254, excluding the number used by the Discovery Camera)
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 192.168.50.1





Camera Connection

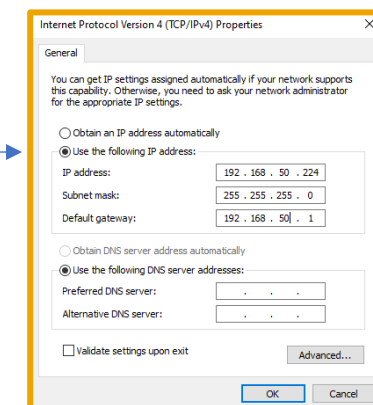
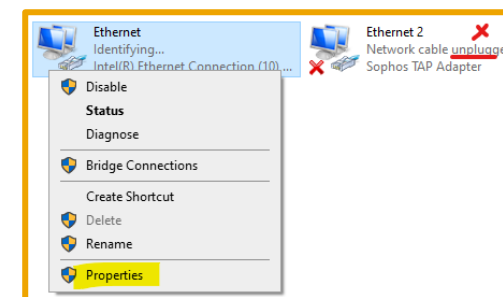
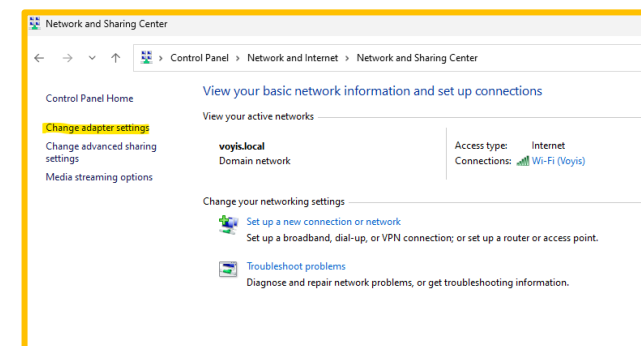
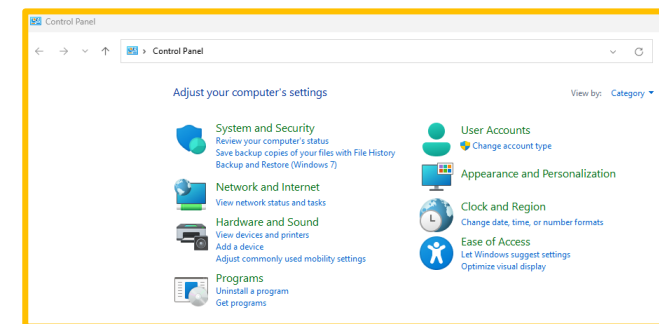
Windows 11

(1) Setup

- Ensure Discovery has 24 VDC power
- Ensure control computer has Ethernet connection to Discovery
- Discovery GUI installed on control computer

(2) IP Settings

- Adjust your computer's subnet to match the Discovery Camera subnet found on the product serial label on camera housing.
 - From the start menu, search “Control Panel”
 - Click on “Network and Internet”
 - Click “Change Adapter Settings”
 - Select “**Properties**” on the Ethernet Connection connected to the Discovery Product.
 - “Use the following IP Address”
 - Product IP Address: 192.168.50.xxx (xxx = any number from 2 to 254, excluding the number used by the Discovery Camera)
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 192.168.50.1

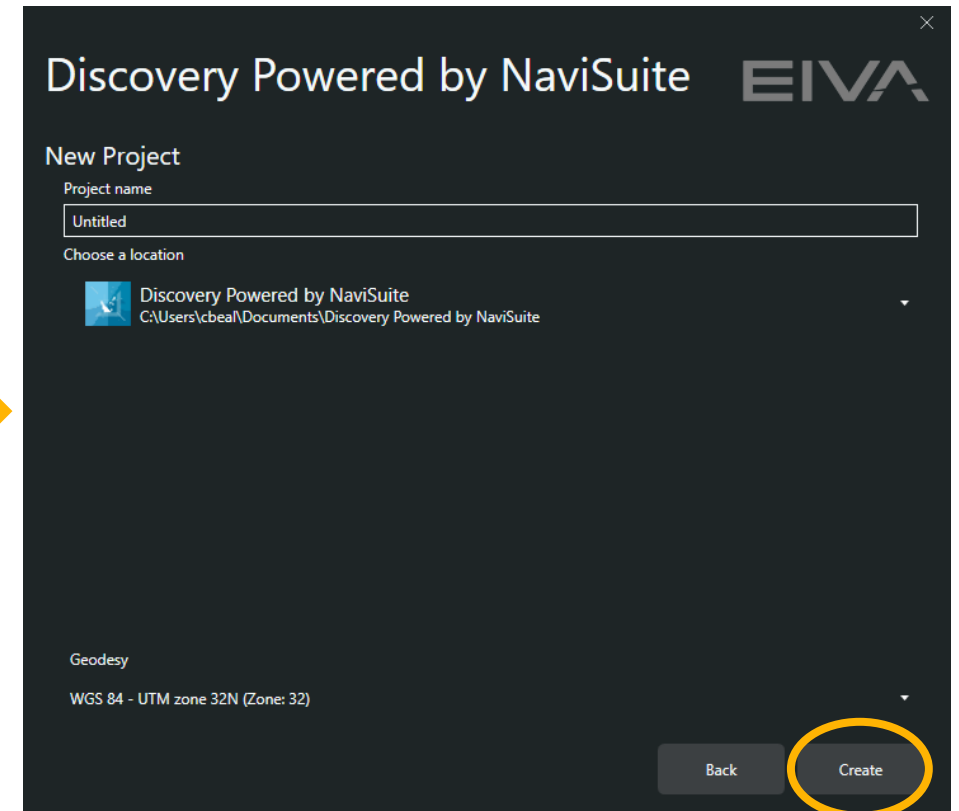
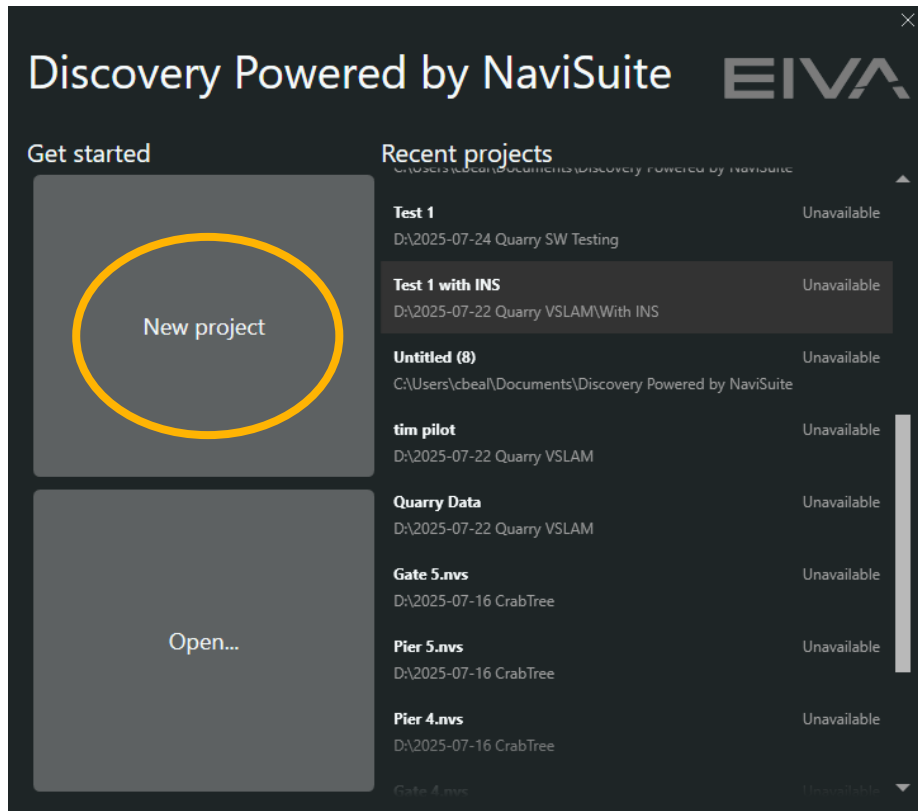




GETTING STARTED

New Project

- When opening the Discovery Powered by Eiva NaviSuite (VSLAM) program you will be prompted to either create a new project or open one
- If you create a new project, you will be prompted to name the project, select the saving location, and select the geodesy

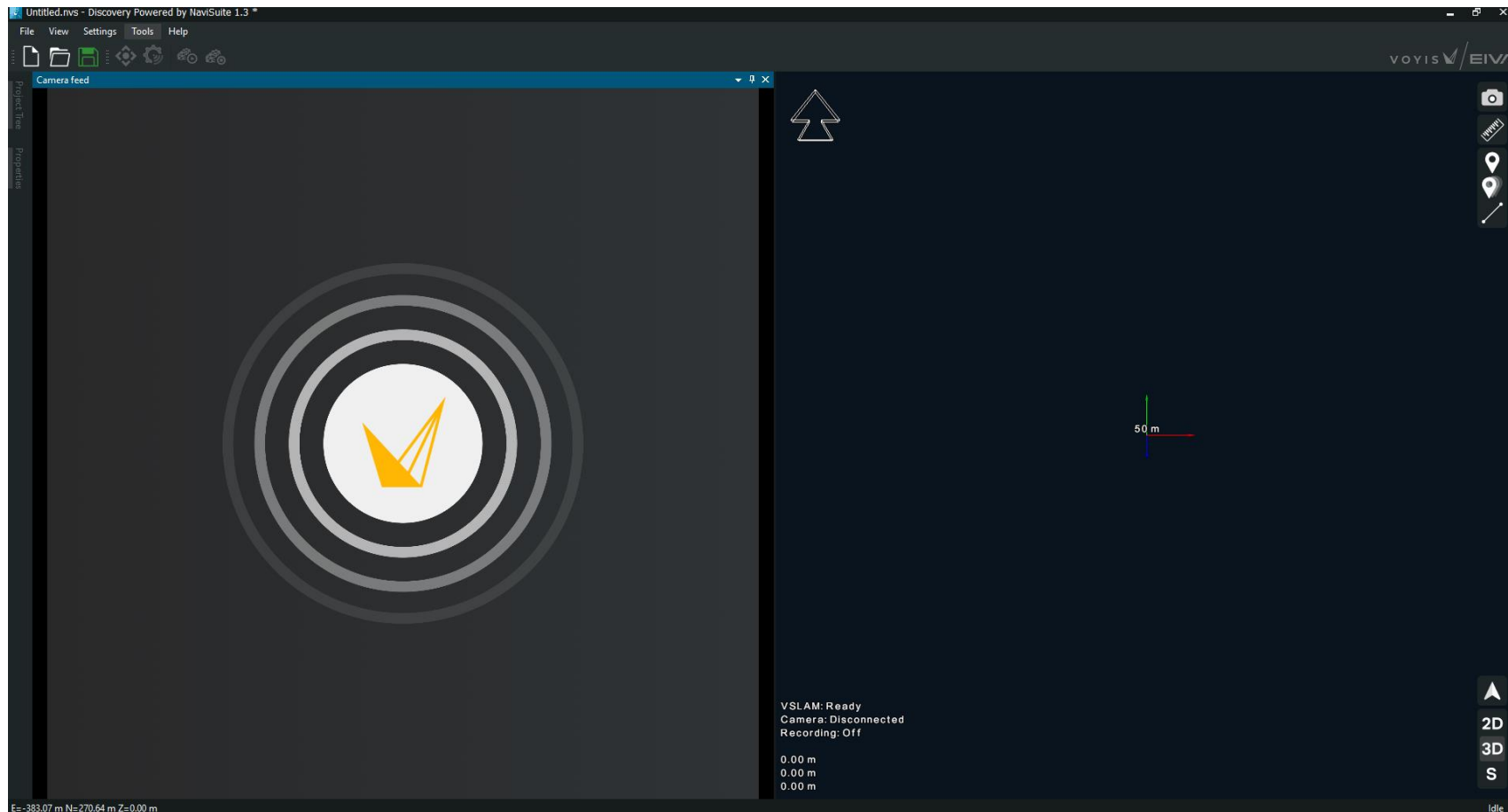




GETTING STARTED

New Project

- Below is the view you will see when you are not connected to a camera. Camera feed on the left and Map view on the right.



- If the camera is connected to the same network, it will automatically connect to the software, and you will see a live video feed from the Voyis Discovery Stereo Camera.



GETTING STARTED

Camera Connection

- You will first be prompted to create a project when you open the software. You can give the project a name of your choice. You can edit a project at any time.

Once connected, you will see the 4 icons below. These 4 buttons are your controls for the camera with the software.

This toolbar consists of:

1. **Camera Control** – Needs to be adjusted before starting VSLAM
2. **Sensor Settings** – Allows user to set up INS
3. **Start VSLAM** – Starts data capture
4. **Stop VSLAM** – Stops data capture





GETTING STARTED

Software Bench Testing

- For testing in air prior to deployment, 3D modelling is limited:
 - The Voyis Discovery Stereo camera is designed and focused **for water**.
 - Video & Images captured in air will be **out of focus** unless 10cm in front of the camera
 - Any air testing will result in unusable 3D data
- Recommended bench testing in air is:
 - Verify the software can connect to the camera
 - Verify that a video feed is active, and you can change settings (video will be out of focus)
 - Check System Status information



GETTING STARTED

Voyis VSLAM SW Compatibility Matrix

The following matrix shows the Software compatibility between different components needed to work between the Discovery camera and VSLAM:

DiscoveryPoweredByNaviSuite	NaviSuiteVSLAM	Voyis Discovery Stereo Camera Firmware	Voyis Discovery GUI	NVIDIA CUDA
Version				
1.00	1.00	1.09	1.1.4	12.1
1.10	1.10	1.09	1.1.4	12.4
1.20	1.20	2.0.1	2.0.1	12.4
1.30	1.30	2.0.1 and/or 2.1.0	2.0.1 and/or 2.1.0	12.4

Controls





HARDWARE CONTROLS

Camera Settings

1
2
3
4



1. Frame Rate (fps) – Video frame rate in frames per second
Recommended: 20 fps

2. Exposure (ms) – Higher exposure results in increased image brightness as more light is received by the sensor

- **Recommended:** Utilize Auto Exposure (not manual)
- If Manual Exposure is used, set based on Lighting Mode:
 - Off – 0.2 ms to 30 ms
 - Strobing – 0.2 ms to 3.5 ms (max strobe is 3.5 ms)
 - Continuous – 0.2 ms to 30 ms
- **Note:** The user should minimize exposure time, while still ensuring a sufficiently illuminated video. A lower exposure will reduce image blur caused by vehicle movement, which improves the results of 3D modeling.

3. Gain (db) – This controls the camera sensor sensitivity

- **Recommended:** Utilize Auto Gain
- 1 dB to 10 dB (>6 dB may cause image graininess)
- **Note:** Gain can be used to increase image brightness without the need to increase exposure which may increase image blur. This is valuable in Strobing mode (max exposure of 3.5 ms), or when using ambient sunlight (Light status: Off)

4. Auto Exposure (ms) – Automatically optimizes camera exposure time to achieve a reasonable image brightness.

- **Recommended:** On. Min & Max depends on Lighting Mode:
 - **Off** – 0.2 ms to 30 ms.
 - **Strobing** – 0.2 ms to 3.5 ms.
 - **Continuous** – 0.2 ms to 20 ms.
- **Note:** If capturing images for 3D modeling, keep the max exposure below 10 ms to reduce blur



Camera Control



5. Auto Gain (dB) – Automatically optimizes camera gain to achieve a reasonable image brightness.

- **Recommended:** On ; 1 dB (min) to 6 dB (max).
- **Note:** Exceeding 6 dB can increase image graininess/noise

6. Auto White Balance – Off by default. Automatically corrects image color in real-time by adjusting colors to balance blue, green, and red channels.

- **Off** – No white balance
- **Dynamic** – Continually adjusted white balance
- **Fixed** – When clicked a one-time white balance adjustment will be captured that will be applied to images. This adjustment will be used unless 'Fixed' is clicked again or another white balance mode is selected
- **Recommended:** Dynamic

7. Toggle Video Streaming – Grants the ability to disable video stream on topside PC

- **Recommended:** ON

8. Aspect Ratio – Adjusts sensor aspect ratio

- **Recommended:** 1.1
- 16:9 allows for slightly higher frame rate (30 fps)

9. Video Resolution – controls the video resolution of the video feed

- **Recommended:** Full Resolution
- **Note:** this only affects video stream, not any data captured
- **Note:** pick Low Latency if you are having bandwidth issues



Camera Control



10

11

12

10. Data Publishing Rate (fps) – The frame rate of stills images captured and saved on the topside computer.

- **Recommended:** 5 fps
- **Note:** This depends on vehicle speed. Higher rates will have more image overlap, but result in more saved data and take up more of the ethernet bandwidth

Though the saving rate can go higher than 5 fps, it is recommended to limit it at 5 fps maximum unless you conduct reliability testing on your network before a survey.

11. Light – controls Mini-Nova LED mode

- **Off** – No light. LEDs do not turn on for image capture. When utilizing ambient sunlight as light source.
 - Requires a higher exposure time, and gain, which can negatively impact image quality
- **Strobing** - ON is the recommended. LEDs flash (strobe) synchronized to camera exposures.
 - This setting “flashes” the lights for up to 3.5 milliseconds
 - This results in the highest possible light output.
- **Continuous** - LEDs continuously on. Reduces light output (1/5th of max)'

12. Camera Onboard Saving – allows for saving data on cameras SSD

- **Location** – Name the folder to save data in
- **Save Raw Data** – Saves raw unprocessed 12-bit Tiff images
 - **Recommended OFF**
- **Save Processed Data** – Saves processed .jpg images

NOTE: Reducing the frame rate below 4 fps will negatively affect the VSLAM tracking and output



HARDWARE CONTROLS

Sensor Settings

Set the INS Input Settings to match your INS system output settings.

Supported Navigation Formats:

- HNAV, PSONNAV, LNAVUTC, HYDROID, DELPHINS, ROVINS, NAVLAB, IXBLUESTD, PHINSSTD, NMEA

- Positional information will be displayed under Navigation Data when the Discovery Stereo system and INS have been properly synced

View the “Voyis x INS Setup” document or Contact support@voyis.com for additional assistance

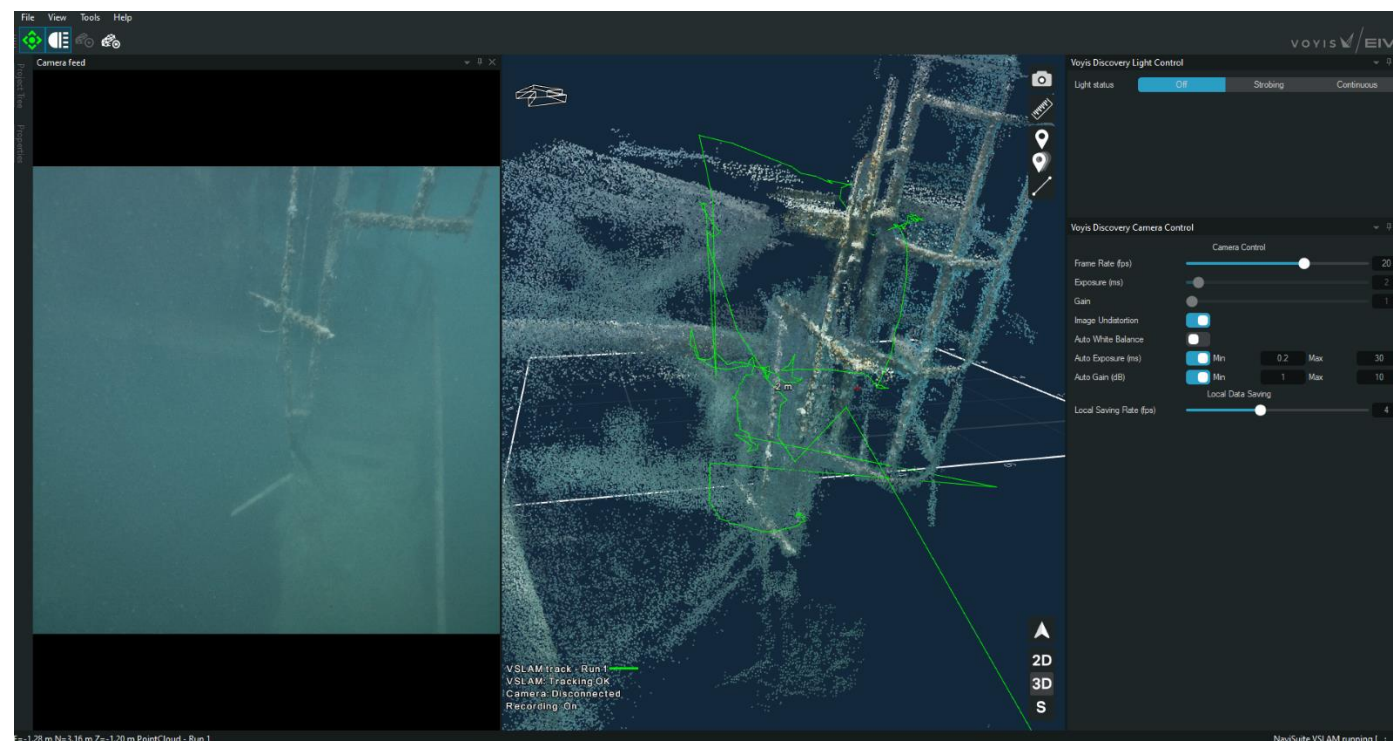


Start VSLAM Acquisition

- Once Settings are selected, you can begin collecting survey data!
- VSLAM is activated (data saving) with the START icon
- Once started, a Point Cloud will begin forming in the **Map View** window, and added to the **Project Tree**
- The Map View will display “Recording On” to indicate data is being collected



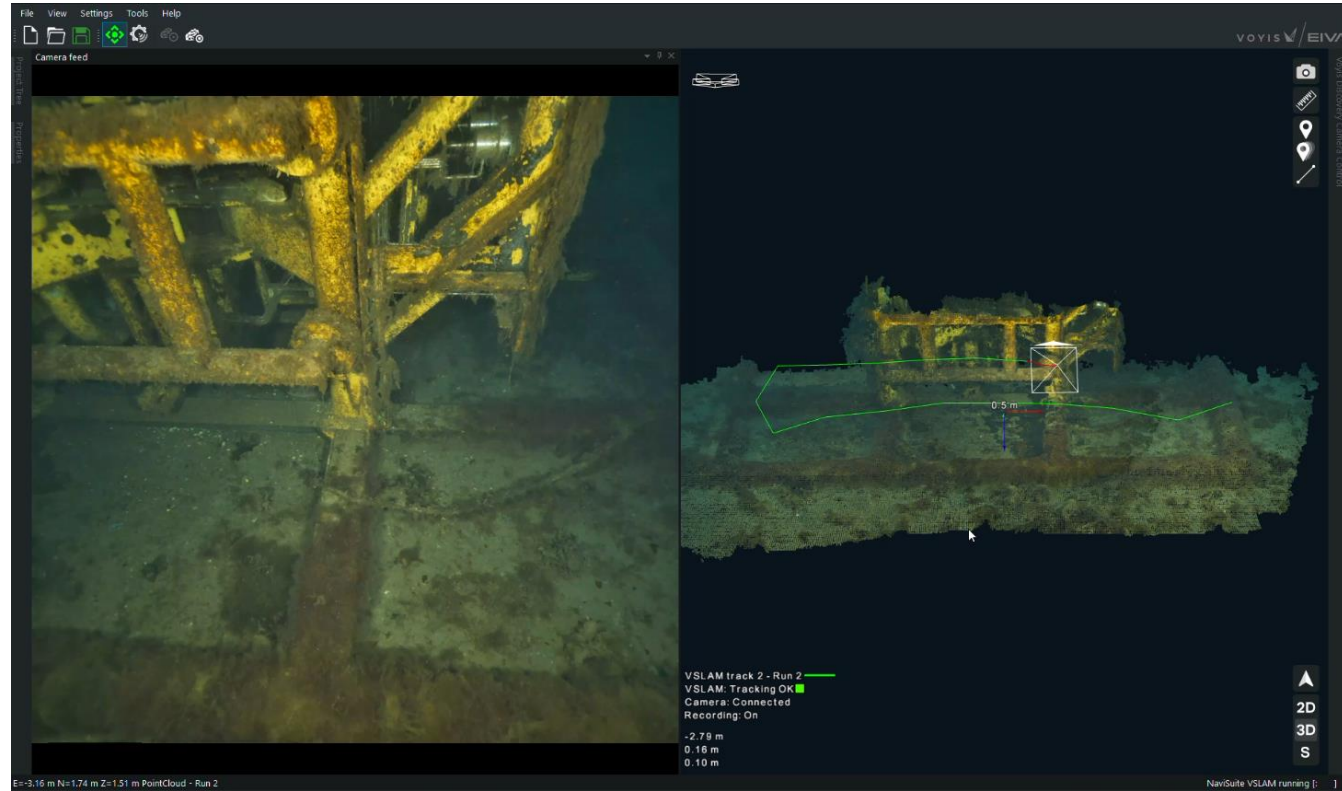
Stop Logging





VSLAM Loop Closure

- VSLAM now displays a “Loop Closure Indicator” at the bottom left of the point cloud screen
 - Green – Indicates a good loop
 - Yellow – Indicates a loop closure could be done to improve point cloud quality
 - Red – Indicates a loop closure should be done as soon as possible to improve point cloud quality
- Loop closures will update the point cloud and adjust the flight path



VSLAM track 3 - Run 3 ———
VSLAM: Tracking OK ■
Camera: Connected
Recording: On

VSLAM track 5 - Run 5 ———
VSLAM: Tracking OK ■
Camera: Connected
Recording: On

VSLAM track 4 - Run 4 ———
VSLAM: Tracking OK ■
Camera: Connected
Recording: On

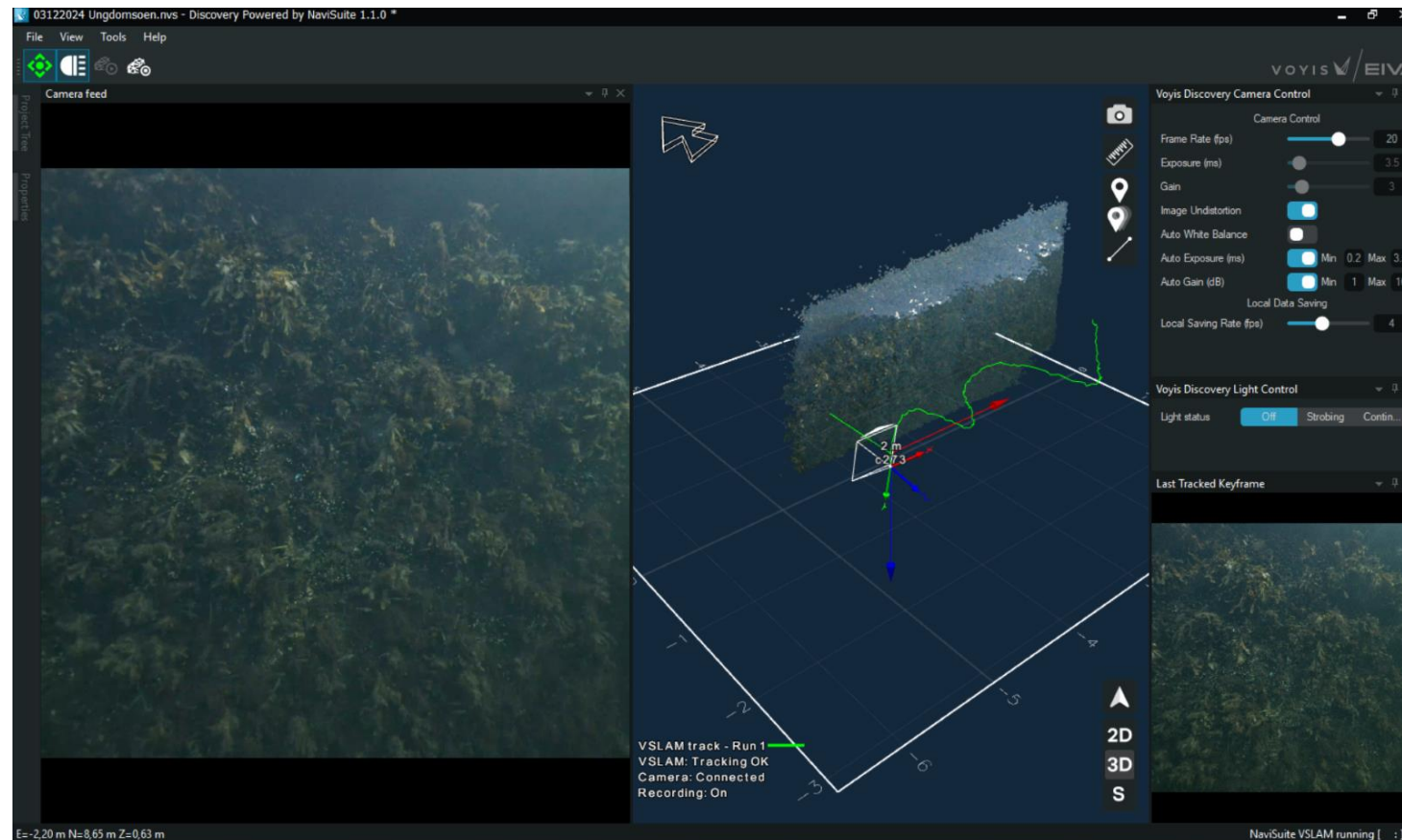


SOFTWARE CONTROLS

Surface Grid

Surface grid can be found nested in the Toolbar or under **Tools menu**.

The surface grid gives a better perspective for the pilot by helping to see relative distance to the object that is modelled during flight. Visualizes the relative distance to the object being modelled





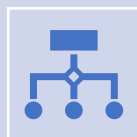
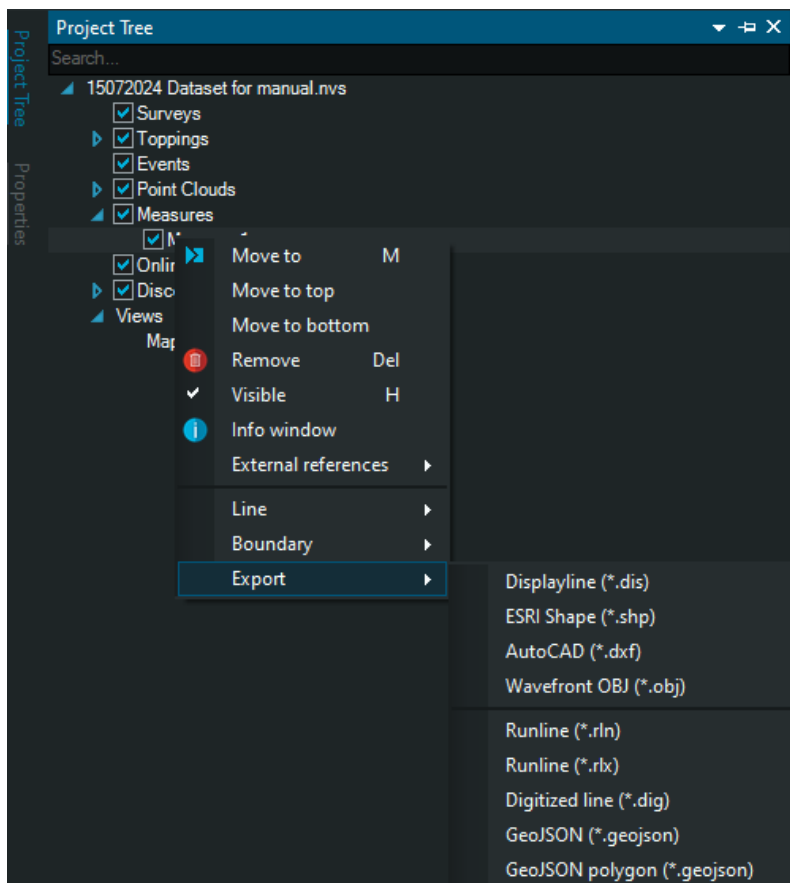
SOFTWARE CONTROLS

Measure Tool

The Measure tool is enabled by pressing the ruler icon in the Map View.

This tool can calculate a distance between points in the model being created within the Map View.

Measure Tool



A new measure object will be created under **Static Objects** in the **Project Tree**. Click a path to get a length at each point, then click **Escape** to end measuring. To remove the measure lines, just select the measure object in the **Project Tree** and press the delete key or delete the object from its context menu.



Measurements are added to the **Project Tree** under **Measures**. Measures can be removed from the project by right-clicking the measure and choosing **Remove**.



Measures can be exported to different formats depending in which program you want to continue quality control and processing of the generated point cloud.

Operating Best Practices





BEST PRACTICES

Operating Requirements

For the software to perform as expected, the following operating conditions should be met:



Ethernet bandwidth must be at least 500 Mbps

Test to ensure it is sufficient with all ROV sensors broadcasting



The camera should be configured with a Saving Rate of 5 FPS.

This ensure sufficient overlap between images.



The camera needs to be oriented towards a structure with visible features to detect & track.

Target features must be visible in the video



The vehicle should be moving at a slow speed (0.2 m/s)



The vehicle needs to move in a stable pattern with a consistent distance to the target

**Optimal Range 1.0m
Min Range 0.5m**



BEST PRACTICES

Turbidity

Water turbidity will directly affect VSLAM functionality. If the software cannot detect features on the target, it cannot generate 3D data points. This may result in:

- Poor quality 3D point cloud
 - Noise in the point cloud
 - A loss in tracking, poor robustness
-
- If there is high turbidity, try to get closer to the target
 - If the lights are too close to the lens of the camera, consider repositioning them further out for less backscatter.

Example of high turbidity, but when close enough the software can detect features





BEST PRACTICES

Target Range

Survey should be conducted as close to the target as possible for best results (0.5 - 1.5m):

- 0.5 - 1.5m provides the best camera focus
- Ensures the highest pixel resolution on the target
- Maximizes the sharpness of target allowing photogrammetry software to easily align images
- Minimizes the effects of turbidity



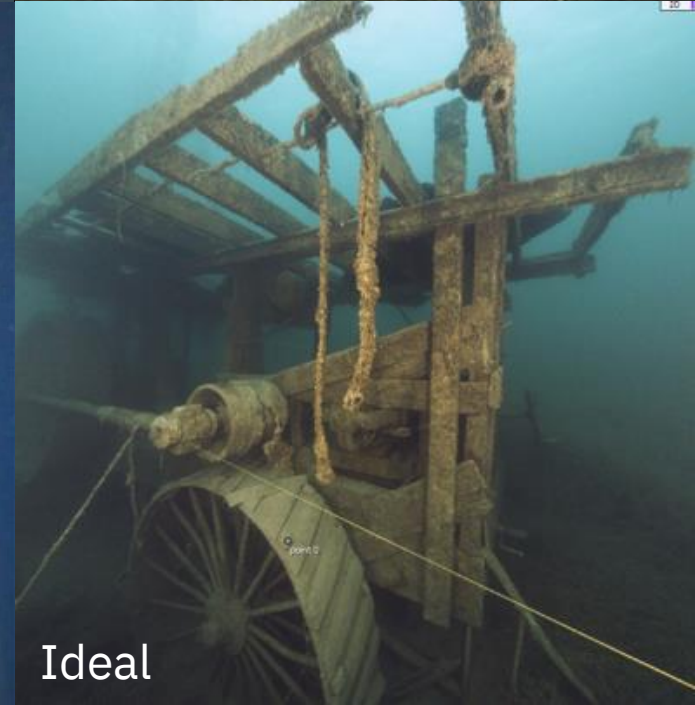
Too Far



Too Far



Ideal



Ideal



BEST PRACTICES

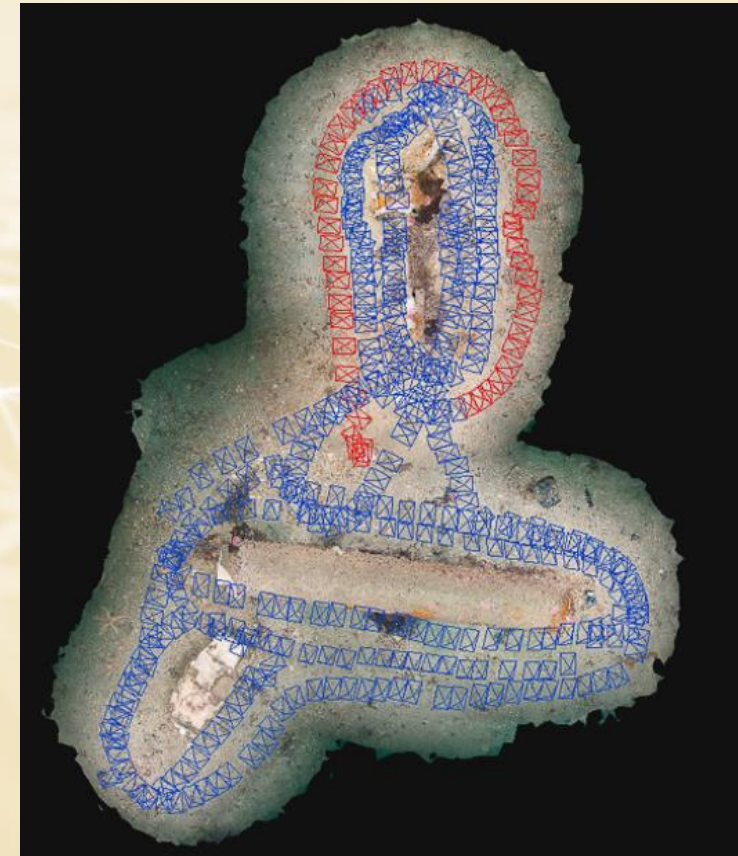
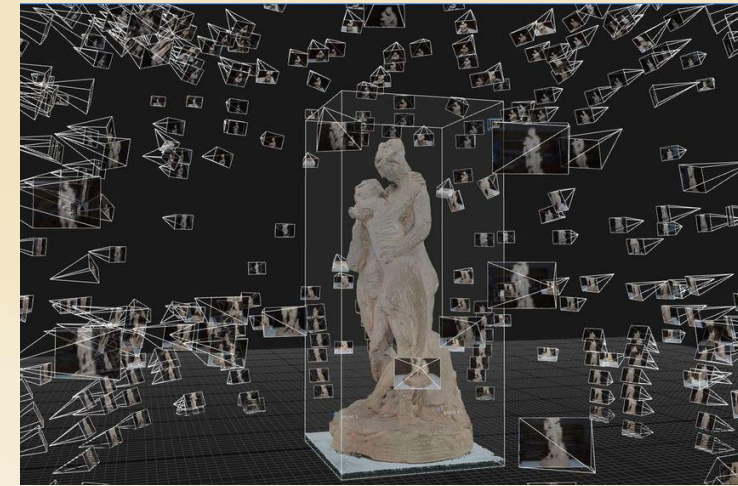
Survey Planning

To generate a high-quality 3D model with photogrammetry, the target must be captured from **many perspectives**:

- Keep the camera pointed perpendicular to the target
- Capture overlapping images in a vertical lawnmower pattern
- Maintain consistent vehicle speed
- Keep the target object in the camera's field of view at all times
- Transitions around corners should be smooth & consistent

ROV Tether management is critical when collecting images with an ROV. Consider your survey route before starting:

- Customers have had better success when laying down their tether on the seabed before beginning the survey.



File Management





File Location

- When you first start up the software you will choose the file path of your data as a new project.
- All data related to this project will be in the project folder at this location

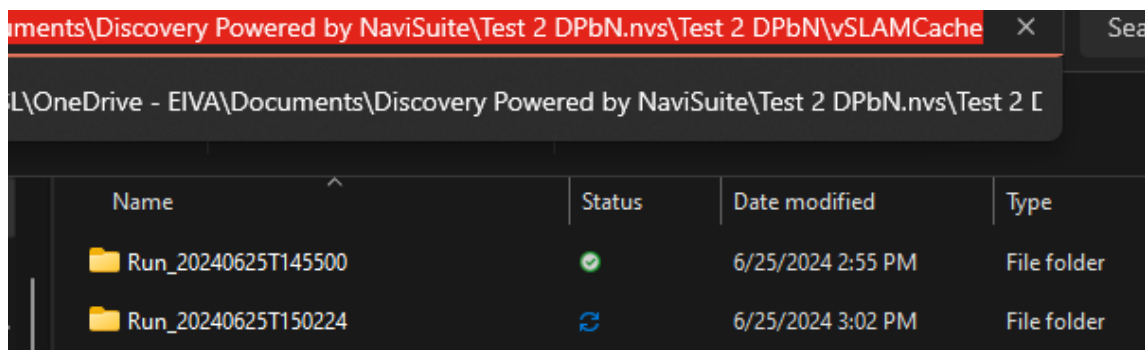


- All **Stills Images** are saved to the folder NaviSuite VSLAMCache into a folder name Run_xxx
 - xxx represents the date and time in yyyyMMdd and the time in 24hr format.



File Location

- The Discovery Powered by NaviSuite folder will contain the Runs from the survey project:



Name	Status	Date modified	Type
Run_20240625T145500	✓	6/25/2024 2:55 PM	File folder
Run_20240625T150224	🔄	6/25/2024 3:02 PM	File folder

- Inside the Run folders will be additional folders from the camera and survey. Click through the folders you see below to navigate to the Left and Right images in which you can use for Photogrammetry Post Processing.

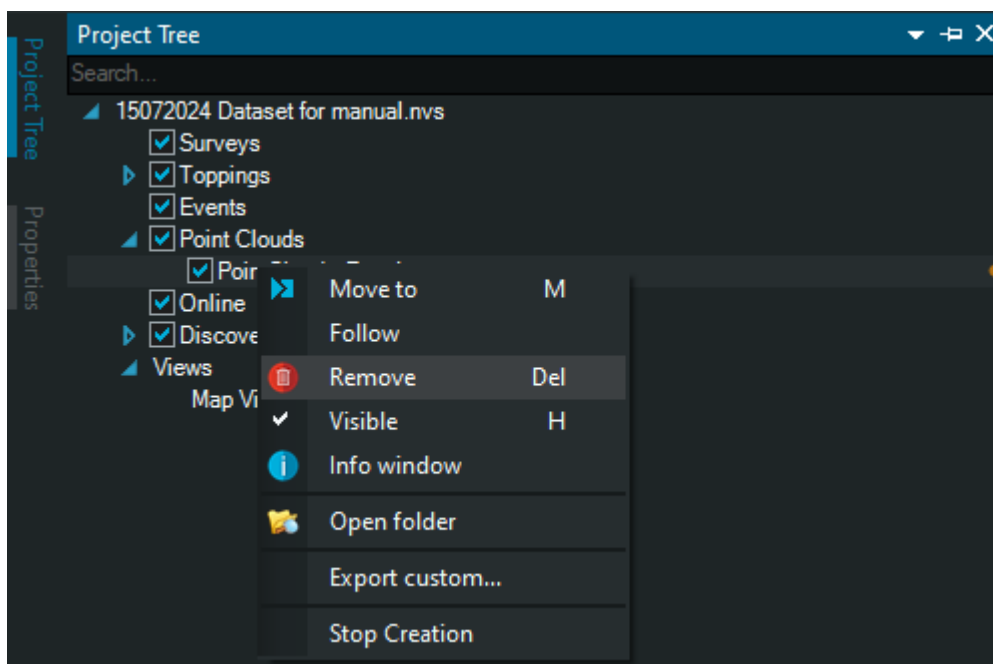
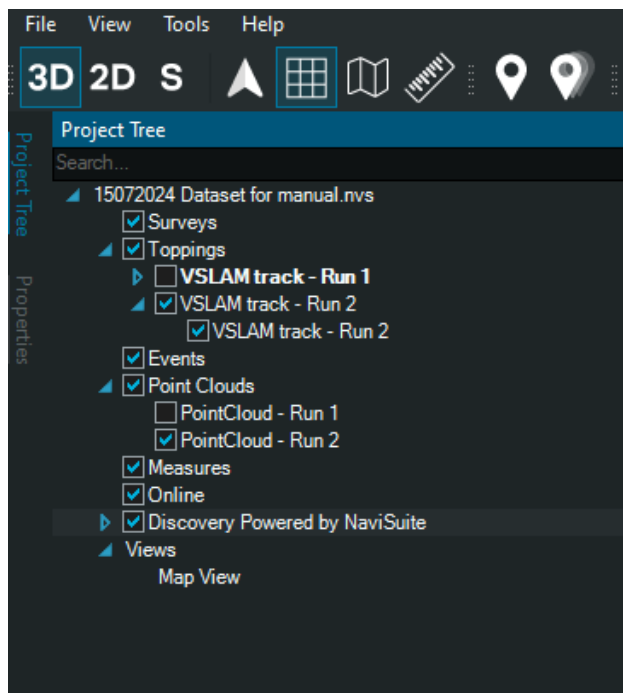




FILE MANAGEMENT

Project Tree

- The Project Tree will include a Topping and Point Cloud from the data collected. These can be individually removed, turned on, or off, within VSLAM without affecting the saved dataset.
- If you wish to start your survey over, a new Run file will be created when you start and stop recording.
- If you wish to remove files so they aren't visible on your screen anymore, this can be done by right-clicking on the file and selecting "Remove"





Export Point Clouds

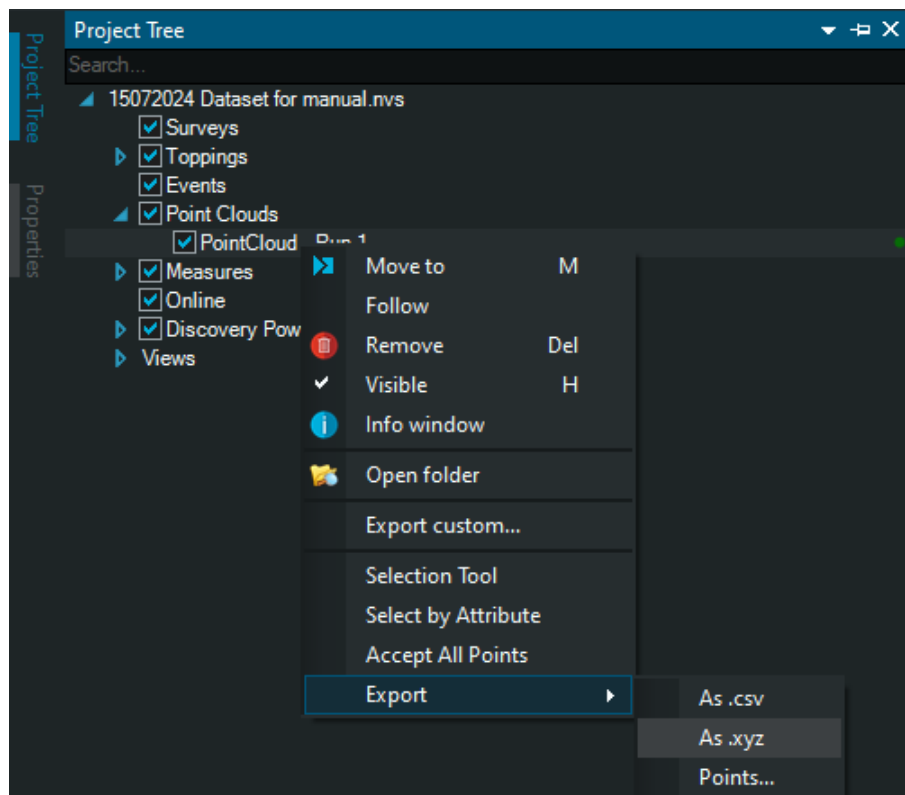
Exporting a point cloud for use in third party software:

- To export a structured point cloud (.xyz format), right-click on the point cloud and select:
 - Export → As .xyz
- To export an unstructured point cloud, right-click on the point cloud and select
 - Export → Points
- To export a structured point cloud (.csv), right-click on the point cloud and select:
 - Export → As .csv

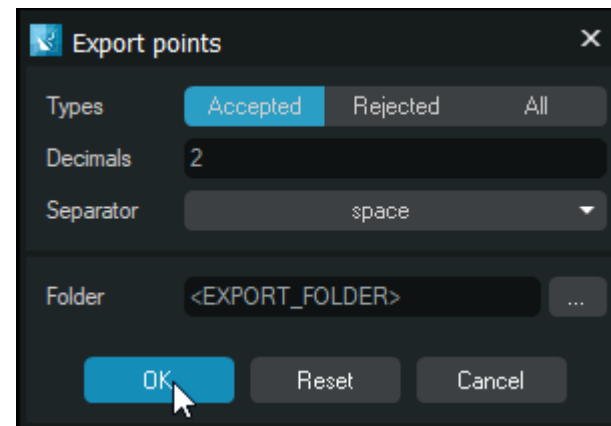


FILE MANAGEMENT

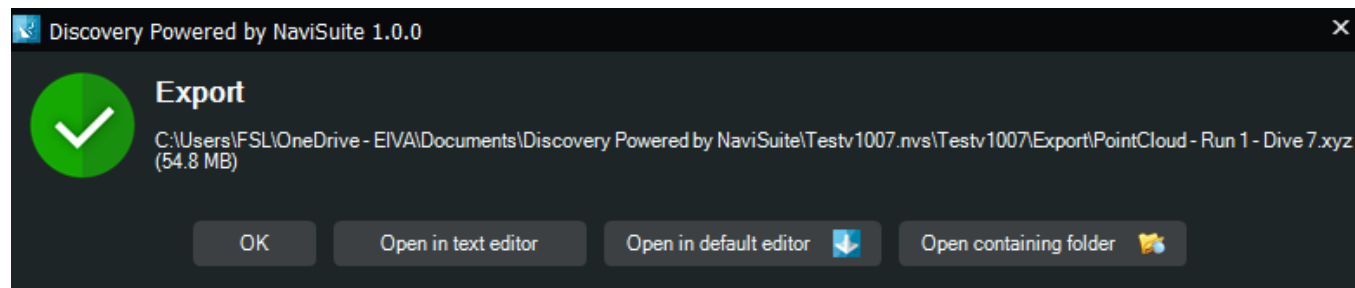
Export Point Clouds



Right Click on the Point Cloud



Choose your export location



View your exported data

Troubleshooting





Camera not connected

The software is running, and camera physically connected, but no data is shown

- Have you check to make sure your computer IP address has been changed to be on the same subnet as the camera?
- Remove all firewalls.
- Open up cmd prompt in windows and ping the camera IP address:
 - Ping 192.168.50.xxx
 - You should get a reply from the camera.
 - If you do not get a reply you will want to check all connections and power and double check the IP address and firewalls. A lot of larger companies will have to cooperate firewalls that will need to be removed in order to operate VSLAM successfully.



TROUBLESHOOTING

Tracking Lost

During a survey, tracking was lost and 3D point cloud stopped generating.

- Is the local frame rate set to 5?
- Are the images of high quality? If not, double-check lighting and exposure settings.
- To re-localise, if tracking is lost, the ROV pilot should follow Operating recommendations.
- Tracking can be lost for numerous reasons, some of them could be that the camera is too close or too far from the object, operating vehicle in unsupported speed...
 - **Best procedure in this case would be to keep the vehicle steady as much as possible and try to find the last tracked keyframe. To enable the Last tracked keyframe please go to Project tree and enable it under active camera.**



TROUBLESHOOTING

3D Model Misalignment

During a survey a second 3D point cloud begins to form, not connected to the previous data (delamination of point cloud)

- Point cloud delamination happens when visiting the same area, but loop closure is not triggered. This occurs when scan lines do not intersect or have low percentage of overlap.
- Generated point cloud will be delaminated, there is visible differentiation between point cloud that is generated. Easiest to recognize delamination is by changing to 2D view and inspecting generated point cloud from the top to see if there is a distinct vertical separation in generated point cloud.
- To avoid or minimize delamination, recommendation is to follow Operating recommendations in Quick start guide.

